

Calculation of low-energy scattering parameters using artificial oscillator trap

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Abstract

We introduce a recipe to estimate the low-energy scattering parameters of a quantum few-body system — scattering length, effective range, and shape parameter — by using only *discrete* state calculations. We place the system in an artificial oscillator trap of varying size and calculate the energies of the resulting discrete states close to the threshold of the system as function of the trap size. The low-energy scattering parameters are then extracted — using a simple analytic formula — from the functional dependence of these energies upon the trap size. We first test the recipe against a simple model problem and then apply it to low-energy nucleon-nucleon scattering within the nuclear Model with Explicit Mesons in one sigma-meson approximation.

1 Introduction

Explicitly Correlated Gaussians is a popular method for numerical calculations of quantum few-body systems [1, 2, 3]. While offering many advantages, the method also has a drawback – it is not *directly* applicable to continuum spectrum calculations due to the specific gaussian asymptotic behaviour of correlated gaussians. Several approaches have been suggested in the literature to remedy this shortcoming, including a class of recipes based on the concept of artificial confining potential, which turns continuum states into discrete quasi-continuum states. The continuum observables are then obtained by analysing the calculated quasi-continuum spectrum using one of many recipes [4, 5, 6, 7, 8, 9, 10, 11, 12, 13].

Here we introduce yet another artificial confining potential method to estimate the low-energy scattering parameters of a few-body system using only discrete spectrum calculations. We place the system in a suitable oscillator trap and calculate the energies of discrete states closest to the threshold of the system as function of the trap size. The low-energy scattering parameters are then extracted from the functional dependence of these energies upon the trap size.

What sets our recipe apart from the existing ones is that i) we explicitly target the low-energy parameters (scattering length, effective range, and shape-parameter) and ii) we extract the low-energy parameters directly from the calculated energies by fitting a simple analytic formula to their functional dependence upon the trap size.

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The recipe is based on the assumption that if two interaction models provide similar energy levels in an artificial trap, their scattering parameters must also be similar. In our recipe one of the interaction models is the zero-range potential model, implemented as a boundary condition at the origin, for which an analytic formula for the spectrum in the oscillator trap can be derived. The low-energy parameters of any other interaction model can then be extracted by fitting its calculated spectrum to the zero-range formula.

We first test the recipe against a simple model — the Volkov potential — and then apply it to calculate the triplet nucleon-nucleon scattering length and effective range within the nuclear Model with Explicit Mesons in one sigma-meson approximation.

2 Harmonic oscillator with zero-range potential

The problem of a quantum harmonic oscillator with a zero-range potential at the origin has been investigated in the past in several publications [11, 12, 14, 15]. Nevertheless, we reproduce here the pertinent formulae for completeness and because previous publications fell short of the shape parameter that we need.

The s -wave Schrödinger equation for a particle with mass m in an oscillator trap with frequency ω and a short-range potential $V(r)$ is given as

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + V(r) + \frac{1}{2}m\omega^2 r^2 - E\right) u(r) = 0, \quad (1)$$

where r is the radial coordinate, $u(r)$ is the radial¹ wavefunction and E is the energy of the particle.

If $V(r)$ is a *zero-range* potential — with scattering length a , effective range r_e and shape parameter P — then it completely disappears from the equation and instead changes the boundary condition at the origin: from the usual condition,

$$u|_{r=0} = 0, \quad (2)$$

to the zero-range boundary condition (see Appendix A),

$$\left.\frac{u'}{u}\right|_{r=0} = \frac{1}{a} + \frac{1}{2}r_e \frac{2mE}{\hbar^2} + Pr_e^3 \left(\frac{2mE}{\hbar^2}\right)^2. \quad (3)$$

The solution to equation (1), regular at infinity, is given by the parabolic cylinder function $U(\alpha, z)$,

$$u(r) \propto U\left(-\frac{E}{\hbar\omega}, \frac{r}{b_o}\right), \quad (4)$$

where $b_o = \sqrt{\hbar/(2m\omega)}$ is the trap range. Inserting this into the boundary condition Eq. (3) gives

$$\frac{1}{b_o} \frac{U'(-\frac{E}{\hbar\omega}, 0)}{U(-\frac{E}{\hbar\omega}, 0)} = \frac{1}{a} + \frac{1}{2} \frac{r_e}{b_o^2} \frac{E}{\hbar\omega} + P \frac{r_e^3}{b_o^4} \left(\frac{E}{\hbar\omega}\right)^2. \quad (5)$$

¹The radial wavefunction $u(r)$ is given as $u(r) = \frac{\psi(r)}{r}$ where $\psi(r)$ is the ordinary wavefunction.

It can be shown (see Appendix B) that

$$\frac{U'(\alpha, 0)}{U(\alpha, 0)} = -\sqrt{2} \frac{\Gamma(\frac{\alpha}{2} + \frac{3}{4})}{\Gamma(\frac{\alpha}{2} + \frac{1}{4})}, \quad (6)$$

which gives the following analytic eigenvalue equation for a particle in a combined zero-range and oscillator potential,

$$-\sqrt{2} \frac{\Gamma(-\frac{E}{2\hbar\omega} + \frac{3}{4})}{\Gamma(-\frac{E}{2\hbar\omega} + \frac{1}{4})} = \frac{b_o}{a} + \frac{1}{2} \frac{r_e}{b_o} \frac{E}{\hbar\omega} + P \left(\frac{r_e}{b_o} \right)^3 \left(\frac{E}{\hbar\omega} \right)^2. \quad (7)$$

For the given a, r_e, P, ω the solutions $E(a, r_e, P, \omega)$ to this equation are the eigen-energies of the corresponding zero-range potential in the oscillator trap with frequency ω and range $b_o = \sqrt{\hbar/(2m\omega)}$. The limit $a \rightarrow 0$ recovers the free s -wave oscillator spectrum $(2n + 3/2)\hbar\omega$.

Alternatively, given an eigen-energy $E(b_o)$ of any interaction model as function of the oscillator range b_o (at least three distinct values), this equation provides the corresponding low-energy scattering parameters a, r_e , and P . And that is the essence of our recipe.

3 Test against simple model

As a test of the introduced recipe we consider a particle in the Volkov potential [16],

$$W(r) = V_R e^{-r^2/b_R^2} + V_A e^{-r^2/b_A^2}, \quad (8)$$

where $V_R = 144.86$ MeV, $b_R = 0.82$ fm, $V_A = -83.34$ MeV, $b_A = 1.60$ fm. The mass of the particle — approximately the reduced mass of two nucleons — is given as [16],

$$m = \frac{1}{2} \frac{(\hbar c)^2}{41.47 \text{ MeV fm}^2} = 469.471 \text{ MeV}. \quad (9)$$

The reported scattering length² for this system is $a = -10.082$ fm and the bound state energy $E_b = -0.54592$ MeV [16].

According to our prescription we now add an artificial oscillator trap with the range b_o to the system. The corresponding Schrödinger equation is then given as

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial r^2} + W(r) + \frac{1}{4} \left(\frac{\hbar^2}{2mb_o^2} \right) \frac{r^2}{b_o^2} \right] u(r) = E u(r). \quad (10)$$

The numerically calculated spectrum of this system for different values of b_o is shown on Fig. (1) as black dots. The energies of the ground state and the first excited state, collected from the last two femtometers, were used to estimate the scattering parameters by fitting to equation Eq. (7). The result is $a = -10.08$ fm, $r_e = 2.38$ fm, and $P = 0.028$.

To illustrate the agreement between the two interaction models (Volkov and zero-range) the figure also shows — with solid lines — the eigen-energies of the zero-range potential with those low-energy parameters. The agreement seems very good and the results seem to have converged within four decimal digits for the scattering length (Fig. (2) illustrates the convergence of the results with respect to the trap size).

²Here we use the opposite sign convention for the scattering length as compared to [16].

Figure 1: Solid dots: the two lowest energy levels of a particle in the Volkov potential and an oscillator trap with range b_o , Eq. (10), as function of the trap range. Solid lines: the corresponding energies of the same particle in the same trap but with the zero-range potential (ZRP), Eq. (7), with $a = 10.08\text{fm}$, $r_e = 2.38\text{fm}$, and $P = 0.028$. The dashed line indicates the bound state energy as calculated in [16].

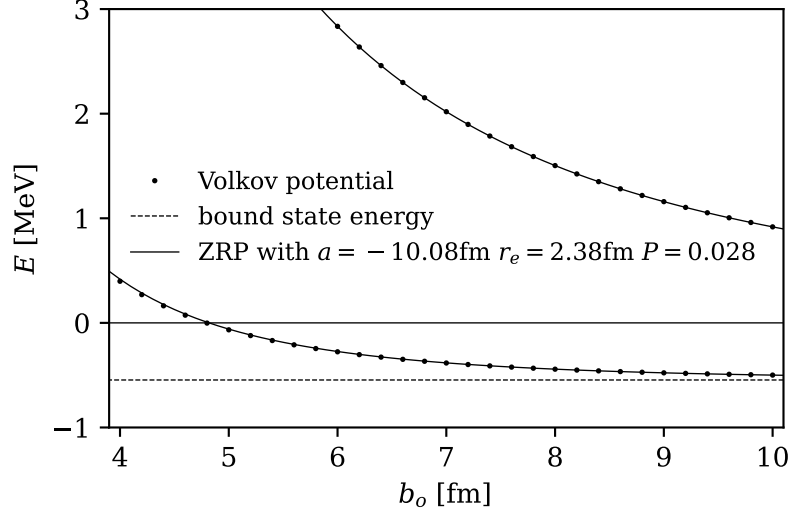
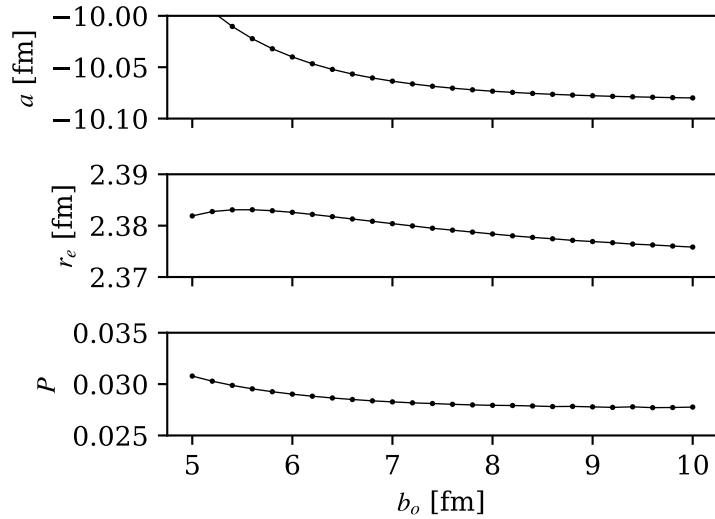


Figure 2: Convergence with respect to trap range: the low-energy parameters of the Volkov potential estimated in a trap with range b_o . The dots are the calculated parameters, the lines are only for guiding the eye.



4 Neutron-proton scattering in σ MEM

In the nuclear model with explicit mesons (MEM) the nucleons interact by (literally) emitting and absorbing mesons [18]. In its simplest version, σ MEM, a pair of nucleons can emit and absorb a (scalar isoscalar) sigma-meson. It has been shown that this version of MEM is able to provide a bound neutron-proton state, the deuteron, with a reasonable binding energy and radius. However the question remains about the corresponding scattering observables – the scattering length and the effective range. In the following we calculate those observables using the introduced prescription.

In σ MEM the physical neutron-proton system is a superposition of two subsystems: the neutron-proton subsystem, and the neutron-proton-sigma subsystem. The total wavefunction, Ψ_{np} , is a two-component structure,

$$\Psi_{np} = \begin{pmatrix} \psi_{np} \\ \psi_{np\sigma} \end{pmatrix}, \quad (11)$$

where ψ_{np} is the wavefunction of the neutron-proton subsystem and $\psi_{np\sigma}$ is the wavefunction of the neutron-proton-sigma subsystem. The corresponding Hamiltonian is a block-matrix operator,

$$H = \begin{pmatrix} K_n + K_p + V_{\text{trap}} & W^\dagger \\ W & K_n + K_p + K_\sigma + m_\sigma \end{pmatrix}, \quad (12)$$

where K_n , K_p , K_σ are kinetic energy operators for the neutron, proton, and the sigma-meson, m_σ is the mass of the σ -meson, V_{trap} is the artificial confining potential, and $W/W^\dagger$ is the sigma-meson generation/annihilation operator taken from [18],

$$W = 20.35\text{MeV} \times \exp\left(-\frac{r_{np}^2 + r_{np\sigma}^2}{(3\text{fm})^2}\right), \quad (13)$$

where r_{np} is the distance between the neutron and the proton, and $r_{np\sigma}$ is the distance between the sigma-meson and the neutron-proton centre of mass.

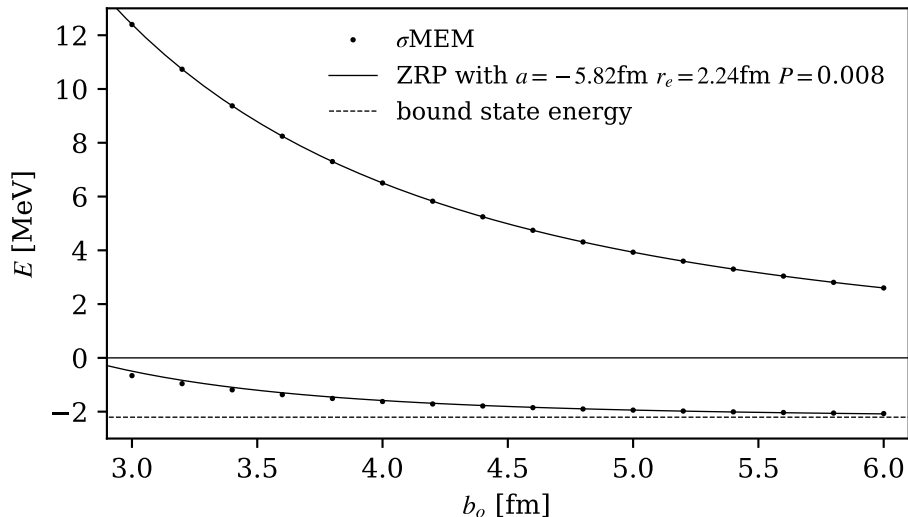
The $np\sigma$ subsystem does not need a trap since at the energies involved—just above the np -threshold—this subsystem is in the classically forbidden region and is therefore confined by itself.

Each component of the total wavefunction is expanded in correlated gaussians, the parameters of which are optimised by minimising the sum of the ground and the first excited state of the system. Experimentations show that as few as 4 gaussians in the np -channel and 12 gaussians in the $np\sigma$ -channel are sufficient to provide the results that are converged to within three decimal digits.

The two lowest calculated eigen-energies of the Hamiltonian Eq. (12) are shown on Fig. (3) together with the energies of a zero-range potential with parameters $a = 5.82\text{fm}$, $r_e = 2.24\text{fm}$, and $P = 0.008$. The convergence of the calculated scattering parameters with respect to trap size is illustrated on Fig. (4).

As in the test case, the zero-range model fits the data very well. However the resulting scattering length and effective range — while qualitatively correct — overestimate the experimental values ($a = 5.41\text{fm}$, $r_e = 1.74\text{fm}$ [19]) indicating that the σ MEM model is not complete. That is expected however as any

Figure 3: Solid dots: the two lowest energy levels of the neutron-proton system in an oscillator trap with range b_o calculated in σ MEM. Solid lines: the corresponding energies of the neutron-proton system in the same oscillator trap interacting via a zero-range potential (ZRP) with the given parameters. The dashed line indicates the binding energy as calculated in [18].



meson exchange interaction model should probably include the pion exchange contribution (which σ MEM lacks).

5 Conclusion

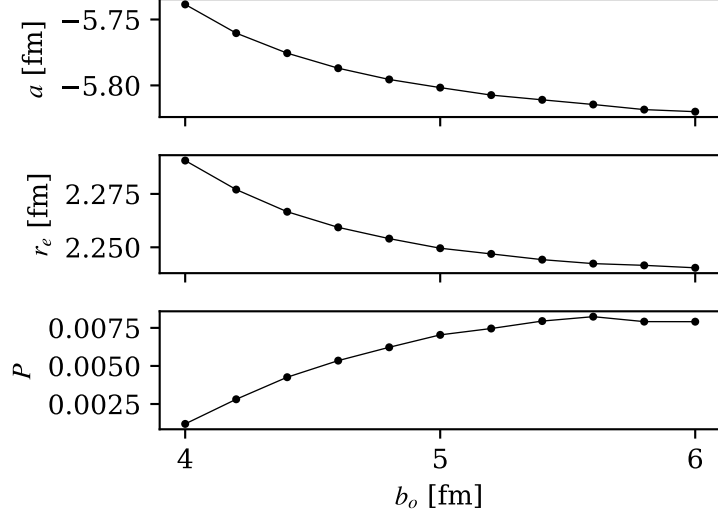
We have introduced a new recipe to estimate the low-energy scattering parameters — scattering length, effective range, and shape-parameter — of a few-body quantum system using only *discrete* spectrum calculations.

The recipe belongs to the class of artificial confining potential recipes. It involves placing the few-body system in a suitable artificial harmonic oscillator trap which turns continuum scattering states into discrete states. The energies of several discrete states close to the threshold of the system are calculated as function of the trap size. The low-energy scattering parameters are then extracted from the functional dependence of the energies upon the trap size.

Unlike the existing recipes from the same class we i) explicitly target the low-energy parameters and ii) extract the parameters directly from the calculated energies by fitting an analytic formula to their functional dependence upon the trap size.

We have first tested the recipe against a simple model and then applied it to calculate the low-energy neutron-proton scattering parameters within the nuclear Model with Explicit Mesons in one sigma-meson approximation. The results show that this approximation — although qualitatively correct — overestimates the scattering parameters by about 10% to 20%, indicating that the approximation is incomplete, most probably due to its lack of the pion exchange

Figure 4: Convergence with respect to the trap range: the σ MEM low-energy neutron-proton parameters estimated in a trap with range b_o . The dots are the calculated parameters, the lines are only for guiding the eye.



contribution. The latter is the subject of the ongoing investigation.

A Zero-range potential as boundary condition

Consider the s -wave radial Schrödinger equation for a particle in a short-range potential $V(r)$,

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial r^2} + V(r) - E \right) u(r) = 0, \quad (14)$$

where r is the radial coordinate, m is the mass of the particle, $u(r)$ is the radial³ wavefunction and E is the energy of the particle. The short-range potential $V(r)$ is assumed to have a certain (short) range b beyond which the potential is negligible. This range divides the space into two regions: the inside region, $r < b$, where the potential is significant, and the outside region, $r > b$, where the potential is negligible. The logarithmic derivatives of the solutions in the inside and outside regions must be matched at $r = b$.

In the outside region, $r > b$, where the potential is negligible, the solution to the equation Eq. (14) is a free wave given as

$$u_{\text{out}}(r) = \sin(kr + \delta), \quad (15)$$

where $\hbar^2 k^2 / (2m) = E$ and δ is the phase-shift. The logarithmic derivative of this solution at $r = b$,

$$\left. \frac{u'_{\text{out}}}{u_{\text{out}}} \right|_{r=b} = k \cot(kb + \delta), \quad (16)$$

³The radial wavefunction $u(r)$ is given as $u(r) = \frac{\psi(r)}{r}$ where $\psi(r)$ is the ordinary wavefunction.

must be matched with the solution from the inside region.

Now, the *zero-range* potential amounts to moving the condition Eq. (16) to the origin,

$$\left. \frac{u'_{\text{out}}}{u_{\text{out}}} \right|_{r=0} = k \cot(\delta). \quad (17)$$

This approximation might be expected to be valid in the low-energy limit where $kb \rightarrow 0$. In the same limit one customarily defines

$$k \cot(\delta) = \frac{1}{a} + \frac{1}{2} r_e k^2 + P r_e^3 k^4 + O(k^6), \quad (18)$$

where a is the scattering length, r_e is the effective range, and P is the shape parameter. Combining the last two equations reduces the zero-range potential to the following energy-dependent boundary condition at the origin,

$$\left. \frac{u'}{u} \right|_{r=0} = \frac{1}{a} + \frac{1}{2} r_e \frac{2mE}{\hbar^2} + P r_e^3 \left(\frac{2mE}{\hbar^2} \right)^2. \quad (19)$$

The zero-range potential can be thought of as a specific limit of a short-range potential where the potential depth tends to infinity while the range approaches zero such that the effective range parameters remain unchanged. Consequently any additional potential that is finite at the origin (in particular the oscillator potential, which is zero at the origin) will not modify the boundary condition for the (infinitely deep and vanishingly short) zero-range potential.

B Parabolic cylinder function

The parabolic cylinder function $U(a, z)$ is the solution to the differential equation

$$\left(-\frac{d^2}{dz^2} + \frac{1}{4}z^2 + a \right) u(z) = 0 \quad (20)$$

with the asymptotics

$$U(a, z) \sim z^{-a-\frac{1}{2}} e^{-\frac{1}{4}z^2}. \quad (21)$$

Its integral representations are given as [20],

$$U(a, z) = \frac{e^{-\frac{1}{4}z^2}}{\Gamma(a + \frac{1}{2})} \int_0^\infty e^{-zt} t^{a-\frac{1}{2}} e^{-\frac{1}{2}t^2} dt, \quad \Re a > -\frac{1}{2}, \quad (22)$$

$$U(a, z) = \sqrt{\frac{2}{\pi}} e^{\frac{1}{4}z^2} \int_0^\infty \cos\left(zt + \frac{\pi}{2}a + \frac{\pi}{4}\right) t^{-a-\frac{1}{2}} e^{-\frac{1}{2}t^2} dt, \quad \Re a < \frac{1}{2}. \quad (23)$$

From these representations one can directly calculate the following values at $z=0$,

$$U(a, 0) = \frac{\sqrt{\pi} 2^{-\frac{a}{2}-\frac{1}{4}}}{\Gamma(\frac{a}{2} + \frac{3}{4})}, \quad U'(a, 0) = -\frac{\sqrt{\pi} 2^{-\frac{a}{2}+\frac{1}{4}}}{\Gamma(\frac{a}{2} + \frac{1}{4})}. \quad (24)$$

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